

Modify System Image

Sub-techniques (2)



Adversaries may make changes to the operating system of embedded network devices to weaken defenses and provide new capabilities for themselves. On such devices, the operating systems are typically monolithic and most of the device functionality and capabilities are contained within a single file.

To change the operating system, the adversary typically only needs to affect this one file, replacing or modifying it. This can either be done live in memory during system runtime for immediate effect, or in storage to implement the change on the next boot of the network device.

ID: T1601

Sub-techniques: [T1601.001](#), [T1601.002](#)

- ① **Tactic:** [Defense Impairment](#)
- ① **Platforms:** Network Devices

Version: 2.0

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[Version Permalink](#)

Procedure Examples

ID	Name	Description
S9013	DRYHOOK	DRYHOOK has modified the Ivanti Connect Secure VPN authentication Perl module <code>DSAuth.pm</code> by reading its contents in the buffer, then finding and replacing select lines of code. ^{[1][2]}

Mitigations

ID	Mitigation	Description
M1046	Boot Integrity	Some vendors of embedded network devices provide cryptographic signing to ensure the integrity of operating system images at boot time. Implement where available, following vendor guidelines. ^[3]
M1045	Code Signing	Many vendors provide digitally signed operating system images to validate the integrity of the software used on their platform. Make use of this feature where possible in order to prevent and/or detect attempts by adversaries to compromise the system image. ^[4]
M1043	Credential Access Protection	Some embedded network devices are capable of storing passwords for local accounts in either plain-text or encrypted formats. Ensure that, where available, local passwords are always encrypted, per vendor recommendations. ^[5]
M1032	Multi-factor Authentication	Use multi-factor authentication for user and privileged accounts. Most embedded network devices support TACACS+ and/or RADIUS. Follow vendor prescribed best practices for hardening access control. ^[6]
M1027	Password Policies	Refer to NIST guidelines when creating password policies. ^[7]
M1026	Privileged Account Management	Restrict administrator accounts to as few individuals as possible, following least privilege principles. Prevent credential overlap across systems of administrator and privileged accounts, particularly between network and non-network platforms, such as servers or endpoints.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0170	Detection Strategy for Modify System Image on Network Devices	AN0482	Defenders may observe adversary attempts to alter or replace a network device's operating system image through anomalous CLI commands, unexpected firmware updates, integrity check failures, or mismatches in version and checksum validation. Suspicious behavior includes modification of image files on storage, OS version output inconsistent with baselines, unexpected reloads or reboots

References

1. [John Wolfram, Josh Murchie, Matt Lin, Daniel Ainsworth, Robert Wallace, Dimitar Andonov, Dhanesh Kizhakkinan, Jacob Thompson. \(2025, January 8\). Ivanti Connect Secure VPN Targeted in New Zero-Day Exploitation. Retrieved April 14, 2026.](#)
2. [Sila Ozeren Hacioglu. \(2025, May 5\). UNC5221's Latest Exploit: Weaponizing CVE-2025-22457 in Ivanti Connect Secure. Retrieved April 13, 2026.](#)
3. [Cisco. \(n.d.\). Cisco IOS Software Integrity Assurance - Secure Boot. Retrieved October 19, 2020.](#)
4. [Cisco. \(n.d.\). Cisco IOS Software Integrity Assurance - Deploy Signed IOS. Retrieved October 21, 2020.](#)
5. [Cisco. \(n.d.\). Cisco IOS Software Integrity Assurance - Credentials Management. Retrieved October 19, 2020.](#)
6. [Cisco. \(n.d.\). Cisco IOS Software Integrity Assurance - TACACS. Retrieved October 19, 2020.](#)
7. [Temoshok, D., et al. \(July 2025\). SP 800-63-4, Digital Identity Guidelines. Retrieved July 29, 2026.](#)